A First Course in Abstract Algebra

Section 45: Unique Factorization Domains

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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45 Unique Factorization Domains

Exercise 45.1: An integer prime

A nonzero nonunit is irreducible when every factorization has a unit factor. If $5 = ab$ in $\mathbb{Z}$, then $|a||b| = 5$. Since 5 is prime, one of $|a|, |b|$ is 1. Thus one factor is a unit, and 5 is irreducible.

Exercise 45.2: A negative prime

The units of $\mathbb{Z}$ are $1, -1$. If $-17 = ab$, then $17 = |a||b|$, so one factor has absolute value 1. Thus -17 is irreducible. Equivalently, it is an associate of the irreducible 17.

Exercise 45.3: A composite integer

The factorization $14 = 2 \cdot 7$ uses two nonunits of $\mathbb{Z}$. Therefore 14 is reducible.

Exercise 45.4: A primitive linear polynomial

Suppose $2x - 3 = gh$ in $\mathbb{Z}[x]$. Degrees add, so one factor, say g, is a nonzero integer c. It divides both 2 and -3, hence $c = \pm 1$. It is a unit. The polynomial is nonconstant, so is itself a nonunit; consequently it is irreducible.

Exercise 45.5: A nonprimitive linear polynomial

In $\mathbb{Z}[x]$, $2x - 10 = 2(x - 5)$. The constant 2 is not a unit of $\mathbb{Z}[x]$, and the nonconstant polynomial $x - 5$ is not a unit either. Therefore the polynomial is reducible.

Exercise 45.6: A linear polynomial over the rationals

Every nonzero constant in $\mathbb{Q}[x]$ is a unit. A factorization of a degree-one polynomial has a degree-zero factor, so that factor is a unit. Thus $2x - 3$ is irreducible in $\mathbb{Q}[x]$.

Exercise 45.7: The same expression in a different domain

Although $2x - 10 = 2(x - 5)$, the factor 2 is a unit in $\mathbb{Q}[x]$. Every factorization has a constant factor because the degree is 1. Therefore $2x - 10$ is irreducible in $\mathbb{Q}[x]$. Irreducibility depends on the coefficient domain.

Exercise 45.8: A linear polynomial modulo eleven

The coefficient 2 is nonzero in the field $\mathbb{Z}_{11}$. Thus $2x - 10$ has degree 1 in $\mathbb{Z}_{11}[x]$ and is irreducible by degree addition. Its constant term is congruent to 1, but this does not change its degree.

Exercise 45.9: Associates in three polynomial rings

Associates are unit multiples. In $\mathbb{Z}[x]$ the only units are ± 1, so the only associates are $2x-7$ and $-2x+7$; four different ones do not exist. In $\mathbb{Q}[x]$, four examples are $2x-7, 4x-14, 6x-21, 8x-28$. In $\mathbb{Z}_{11}[x]$ the same four unit multiples are distinct after reducing coefficients: $2x+4, 4x+8, 6x+1, 8x+5$. Distinctness follows also by comparing their leading coefficients.

Exercise 45.10: Factorization over three domains

First extract the integer content:

$$4x^2 - 4x + 8 = 2 \cdot 2\,(x^2 - x + 2).$$

The discriminant of $x^2 - x + 2$ is $1 - 8 = -7$, so it has no rational root. A quadratic over a field is irreducible exactly when it has no root. Gauss's lemma therefore makes this primitive polynomial irreducible in $\mathbb{Z}[x]$ as well. This is the factorization into irreducibles over $\mathbb{Z}$. Over $\mathbb{Q}$, 4 is a unit, and $4(x^2 - x + 2)$ is an irreducible polynomial times a unit. Modulo 11, the roots of $x^2 - x + 2$ are 5 and 7, because $5 + 7 = 1$ and $5 \cdot 7 = 2$ modulo 11. Thus

$$4x^2 - 4x + 8 = 4(x - 5)(x - 7) \quad \text{in } \mathbb{Z}_{11}[x].$$

The two linear factors are irreducible and 4 is a unit.

Exercise 45.11: All gcds of three integers

Euclidean division gives $3250 = 13(234) + 208$, $234 = 208 + 26$, and $208 = 8(26)$. Also $1690 = 65(26)$. Thus the positive gcd is 26. Since the only units of $\mathbb{Z}$ are ± 1, all gcds are 26 and -26.

Exercise 45.12: All gcds with a negative entry

We may replace -1960 by 1960 without changing divisibility. The calculations $1960 = 2(784) + 392$, $784 = 2(392)$, $448 = 392 + 56$, and $392 = 7(56)$ give positive gcd 56. All gcds are $56, -56$.

Exercise 45.13: All gcds of four integers

The divisions $2178 = 5(396) + 198$ and $396 = 2(198)$ give $\gcd(2178, 396) = 198$. Since $792 = 4(198)$ and $594 = 3(198)$, the remaining entries do not change it. The gcds are $198, -198$.

Exercise 45.14: Integer content

The positive gcd of $18, -12, 48$ is 6. Consequently $18x^2 - 12x + 48 = 6(3x^2 - 2x + 8)$. The coefficients $3, -2, 8$ have gcd 1, so the second factor is primitive. Changing the sign of both factors gives the equivalent associate choice.

Exercise 45.15: Content over a field

In $\mathbb{Q}$ every nonzero element is a unit and is a gcd of these coefficients. One valid choice is content 1 and primitive factor $18x^2 - 12x + 48$ itself. For comparison, the choice $6(3x^2 - 2x + 8)$ is also valid. Over a field every nonzero polynomial is primitive: every common nonzero divisor of its coefficients is already a unit.

Exercise 45.16: A polynomial already primitive

The coefficients $2, -3, 6$ have gcd 1, since any common divisor of 2 and 3 is a unit. Thus $2x^2 - 3x + 6 = 1(2x^2 - 3x + 6)$ is the required decomposition in $\mathbb{Z}[x]$.

Exercise 45.17: Content modulo seven

The coefficient ring is the field $\mathbb{Z}_7$. Take content 1 and primitive factor $2x^2 + 4x + 6$. Alternatively choose content 2 and write $2(x^2 + 2x + 3)$, since $2^{-1} = 4$ modulo 7. All nonzero constants are units, so these choices are equivalent.

Exercise 45.18: Definition of associates

The version using a/b is not suitable when $b = 0$. The correct definition for all $a, b \in D$ is: a and b are associates if $a = ub$ for some unit $u \in D$. This includes 0 being an associate of itself. If $b \neq 0$, the quotient criterion in the fraction field is equivalent, provided that quotient is a unit of D.

Exercise 45.19: Definition of irreducible

An irreducible is a nonzero nonunit $p \in D$ such that whenever $p = ab$ with $a, b \in D$, at least one of a, b is a unit. One cannot prohibit every factorization, because $p = 1 \cdot p$ is always available.

Exercise 45.20: Definition of prime

A prime is a nonzero nonunit p such that $p \mid ab$ implies $p \mid a$ or $p \mid b$, for every $a, b \in D$. The expression "smaller" has no intrinsic meaning in an arbitrary integral domain.

Exercise 45.21: Ten statements about factorization

(a) True. In a field there are no nonzero nonunits, so the factorization requirement is vacuous.

(b) True. The only ideals of a field are (0) and (1).

(c) True. The PID factorization theorem proves existence and uniqueness.

(d) False. $\mathbb{Q}[x, y]$ is a UFD, but (x, y) is not principal: a common divisor of x, y is a unit, which would make that ideal the whole ring.

(e) True, by Gauss's theorem applied to the UFD $\mathbb{Z}$.

(f) False. The irreducibles 2 and 3 of $\mathbb{Z}$ are not associates.

(g) False. $\mathbb{Z}$ is a PID but $(2, x)$ is not principal in $\mathbb{Z}[x]$.

(h) True, by the polynomial UFD theorem.

(i) False literally. Associates may replace p: $2 \mid 6$, but $6 = (-2)(-3)$ has no factor literally equal to 2. The correct claim is that an associate of p appears.

(j) True. Being an integral domain is part of the definition of UFD.

Exercise 45.22: Irreducibles in a polynomial UFD

There are two types. Constant irreducibles are precisely the irreducibles of D. A nonconstant $f \in D[x]$ is irreducible precisely when it is primitive and irreducible in $F[x]$. Indeed, a nonunit content would itself give a factorization; after removing content, Gauss's lemma transfers reducibility between $D[x]$ and $F[x]$. Constant nonzero elements become units in $F[x]$, so they must be treated separately.

Exercise 45.23: Irreducible after extending coefficients

Take $D = \mathbb{Z}$, $F = \mathbb{Q}$, and $g(x) = 2x + 2$. It is degree 1, hence irreducible in $\mathbb{Q}[x]$. In $\mathbb{Z}[x]$ the factorization $g = 2(x + 1)$ has two nonunit factors, so it is reducible. Its nonunit content is precisely what disappears on passing to F.

Exercise 45.24: Why zero divisors obstruct factorization

The element $(1, 0)$ is a nonzero nonunit of $\mathbb{Z} \times \mathbb{Z}$, but $(1, 0) = (1, 0)(1, 0)$, a product of two nonunits. More strongly, it cannot be a product of irreducibles. Suppose $(1, 0) = \prod_j (a_j, b_j)$. All a_j are ± 1 and at least one $b_j = 0$. That factor $(\pm 1, 0)$ is reducible because $(\pm 1, 0) = (1, 0)(\pm 1, 0)$. Hence there is no irreducible factorization of $(1, 0)$. The domain hypothesis prevents precisely this kind of idempotent behavior.

Exercise 45.25: A prime is irreducible

Let p be prime and write $p = ab$. Since $p \mid ab$, assume, after exchanging factors if necessary, that $p \mid a$. Write $a = pc$. Then $p = pcb$; cancel the nonzero p in the domain to obtain $1 = cb$. Thus b is a unit. Since a prime is by definition a nonzero nonunit, this proves irreducibility.

Exercise 45.26: An irreducible in a UFD is prime

Suppose $p \mid ab$, so $ab = pc$. If a or b is zero, p divides that factor. Otherwise factor a, b, c into irreducibles, allowing a unit factor when necessary. The concatenated factorizations of a, b and the factorization p followed by that of c represent the same nonzero element. Uniqueness forces an associate of p to occur in the factorization of a or b. It therefore divides that factor, as does p.

Exercise 45.27: Associates form an equivalence relation

Reflexivity follows from $a = 1a$. If $a = ub$ with u a unit, then $b = u^{-1}a$, proving symmetry. If additionally $b = vc$ with v a unit, then $a = (uv)c$, and uv is a unit, proving transitivity. These arguments need no cancellation and work even in a commutative ring with zero divisors.

Exercise 45.28: Products of nonzero nonunits

If a, b are nonzero nonunits, the domain condition gives $ab \neq 0$. If ab were a unit with inverse c, then $a(bc) = 1$ and $b(ac) = 1$, making both factors units, a contradiction. Thus the set is closed. It is not a group: an identity on this nonempty set would satisfy $ea = a$ for nonzero a, hence $e = 1$ by cancellation, but 1 is excluded. If the domain is a field, the set is empty and is still not a group.

Exercise 45.29: A divisor of a primitive polynomial

Write the primitive polynomial as $f = gh$ in $D[x]$. If a nonunit irreducible p divided every coefficient of g, then it would divide every coefficient of $gh = f$. That contradicts the content of f being a unit. Since every nonunit of the UFD has an irreducible divisor, the coefficients of g have no common nonunit divisor; hence g is primitive. The same argument applies to any divisor, constant or nonconstant.

Exercise 45.30: Containment in a maximal ideal

The statement needs the word *proper*: the ideal D itself is not contained in a maximal ideal. For a proper ideal I in a PID, suppose no maximal ideal contains it. Then I is not maximal, so there is a proper ideal I_1 strictly containing I. Repeating produces $I \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$. The ascending chain condition for a PID forbids this. Thus the process terminates at a maximal ideal containing I.

Exercise 45.31: Factoring a difference of cubes

We have $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. Regard these as polynomials in x over $\mathbb{Q}[y]$. The first is primitive and degree 1, so is irreducible over $\mathbb{Q}(y)$ and therefore over $\mathbb{Q}[y]$. The second is primitive and has discriminant $-3y^2$. This is not a square in $\mathbb{Q}(y)$: otherwise dividing a putative square root by y would give a rational function whose square is -3, impossible in the ordered field $\mathbb{Q}(y)$ (or by comparing coprime numerator and denominator). Thus this quadratic is irreducible over $\mathbb{Q}(y)$; Gauss's lemma gives irreducibility in $\mathbb{Q}[x, y]$.

Exercise 45.32: Three equivalent finiteness conditions

Plan. Prove a cycle of implications for ideals ordered by inclusion.

ACC implies the maximum condition.

If a nonempty collection has no maximal member, choose an ideal in it and then a strictly larger member repeatedly. This contradicts ACC.

The maximum condition implies finite generation.

For an ideal I, consider all ideals generated by finite subsets of I (including the empty generating set). A maximal such ideal J exists. If $a \in I \setminus J$, adjoining a gives a strictly larger finitely generated ideal inside I, a contradiction. Thus $J = I$.

Finite generation implies ACC.

Given $I_1 \subseteq I_2 \subseteq \cdots$, its union I is an ideal: any two elements lie together in a sufficiently late term, and multiplication by ring elements remains in that term. A finite generating set of I lies in some I_N, so $I = I_N$ and all later terms coincide with it. This is ACC.

Exercise 45.33: Descending chains and minimal ideals

As usual the minimum condition must concern *nonempty* collections; an empty collection cannot contain a minimal member. If DCC holds and a nonempty collection had no minimal member, successively choosing a smaller member would produce an infinite strictly descending chain. Conversely, apply the minimum condition to the set of terms of a descending chain. If I_N is minimal among those terms, then for $n \geq N$, $I_n \subseteq I_N$ forces $I_n = I_N$. Thus the chain stabilizes and DCC holds.

Exercise 45.34: ACC without DCC

Use $\mathbb{Z}$. Every ideal is $n\mathbb{Z}$, hence finitely generated, so ACC holds by Exercise 45.32. But

$$2\mathbb{Z} \supsetneq 4\mathbb{Z} \supsetneq 8\mathbb{Z} \supsetneq \cdots$$

is an infinite strictly descending chain: 2^k belongs to the kth ideal but not the next. Therefore DCC fails.